Ling 285
Human Language and Technology
Lecture 8

Dynamical Systems 6

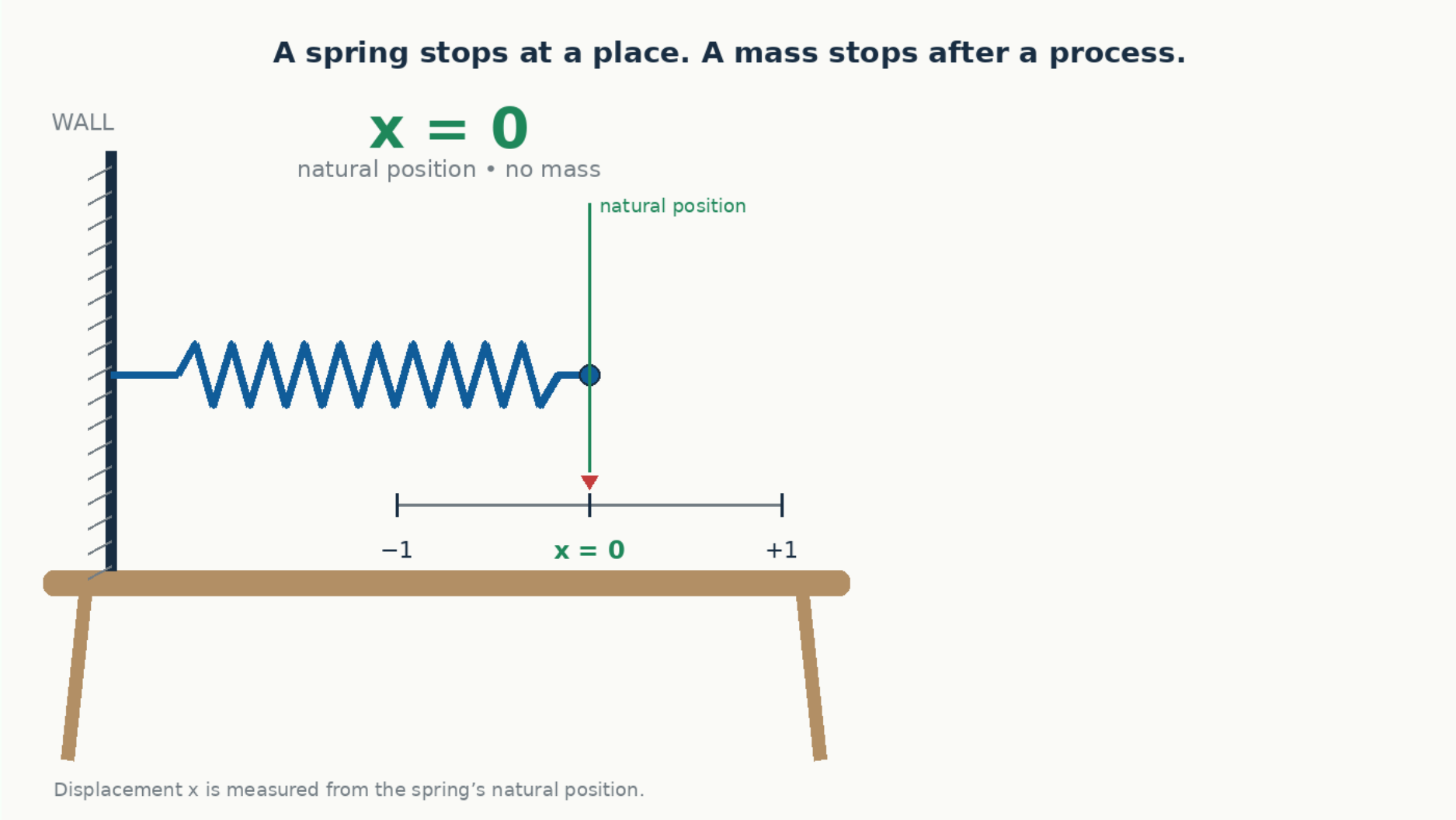
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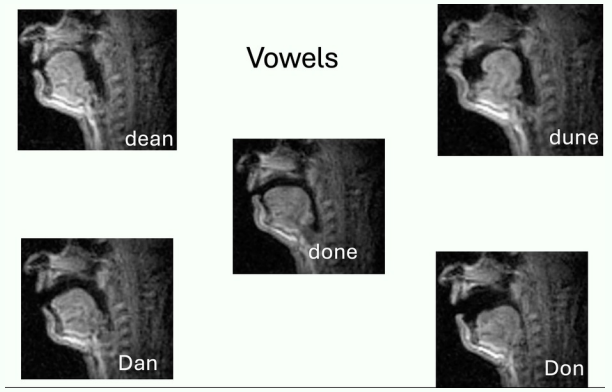
First Exam in 2 weeks

Spring, thenMass-Spring System



Perturbation Theory

- If we increase the mass m of a mass-spring system, will the frequency increase or decrease?
- Frequency will Decrease
- If we increase the stiffness k?
- Frquency will Increase
- Perturbation Theory:
  - m ↑ f↓
  - k ↑ f↑

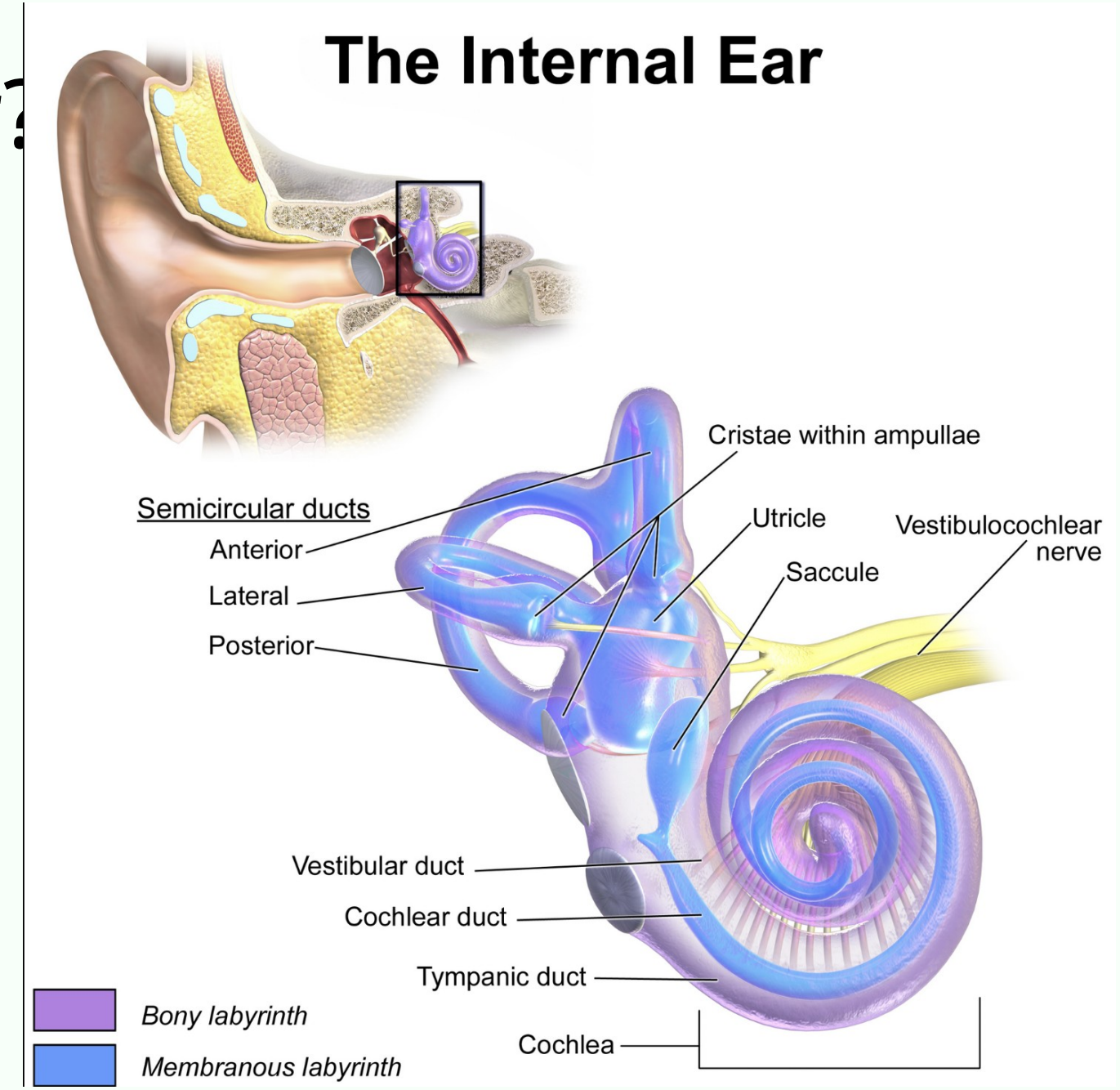


- When we speak, we change the effective mass and stiffness of different portions of air in the vocal tract, differently for each vowel and consonants, so when the air vibrates (due to vocal fold vibration), it vibrates at many frequencies, specific to that V or C, depending on where and how much the air is squeezed!!!!
- Somehow, for you to hear and interpret the speech, I'm producing, your ear needs to figure out what are the low and mid and high frequencies in my voice are, and for the brain to tell, I've just heard a high frequency of this much and a low frequency of this much, so he said, e.g., scene, vs. two other frequencies, therefore he said "son". How?

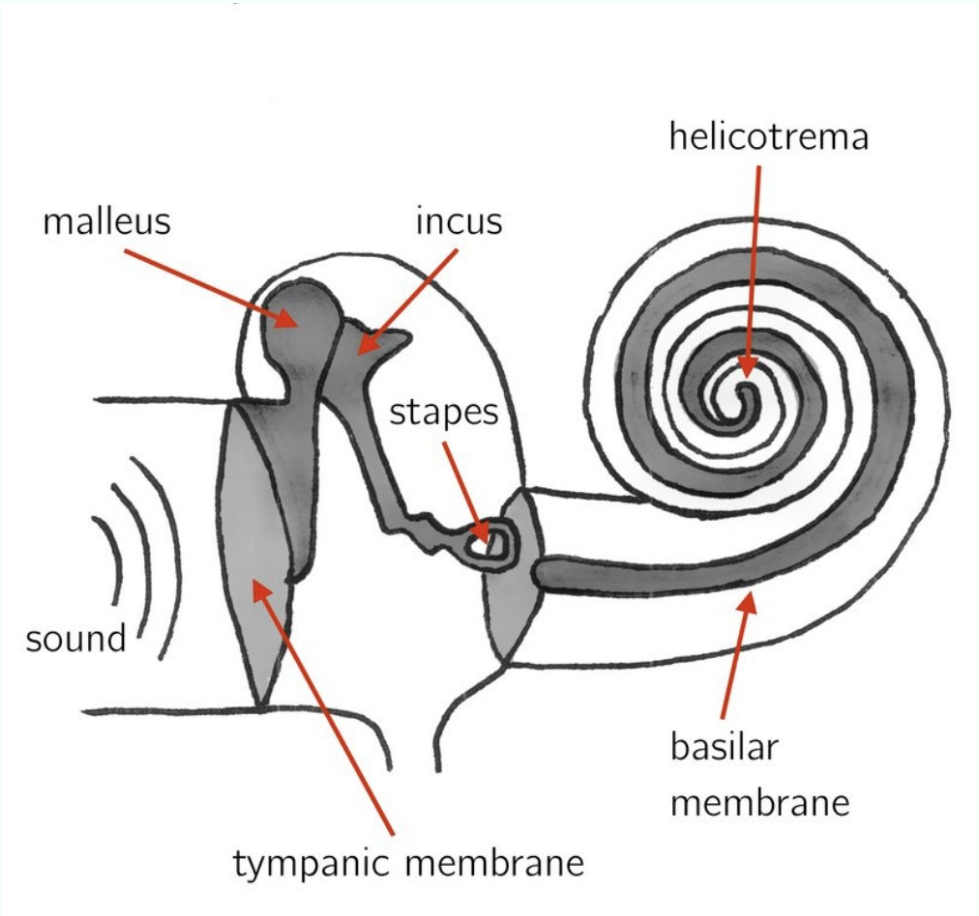
How do you hear?

You don’t need to memorize any of these names, except cochlea

Waves of vibration in the air arrive at your external ear, then travel inward to your inner ear, a part called the cochlea, the part that sends signals to your brain that say what frequencies are in the sound.



Basilar membrane



Hearing Explained!

Frequenc
y



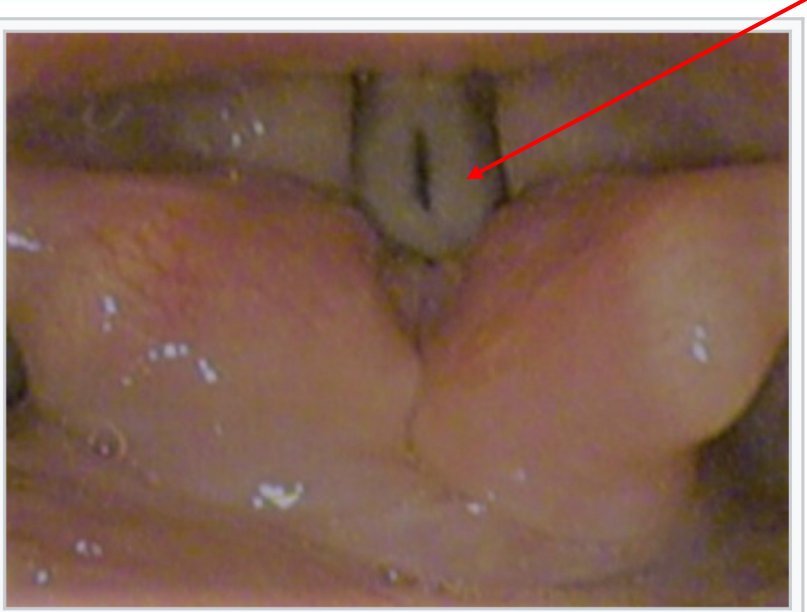
Georg von Bekesy Nobel prize in 1961

Neuron
s

To
brain

Vocal folds

- Your vocal folds are muscles that are about 1 cm in length and 1 mm in height.
- Muscle has mass and stiffness
- Using about 10 other muscles hooked to the bones/cartilages of the housing of the folds, we can stretch them making them stiffer or bunching them making them effectively heavier.
- Why?
  - Singing!
  - In English: Conveying questions, mistrust, degree of belief, surprise, emotion, etc.!
  - In tone language: meanings of words!!!!
- We call the vibration frequency of vocal folds pitch, F0, fundamental frequency.



Vocal folds (speaking)

How/Why do they vibrate?

If we take air into our lungs, and close the vocal folds, a lot of pressure builds beneath them, and if the pressure is high enough, it blows them apart, and there is airflow between them. But just like a curtain in a shower, the airflow sucks the vocal folds back in, and the air pressure below blows them apart again, and this repeats 100's of time per second. The suction force is called

Harmonic Oscillator

$$\dot{x} = v$$

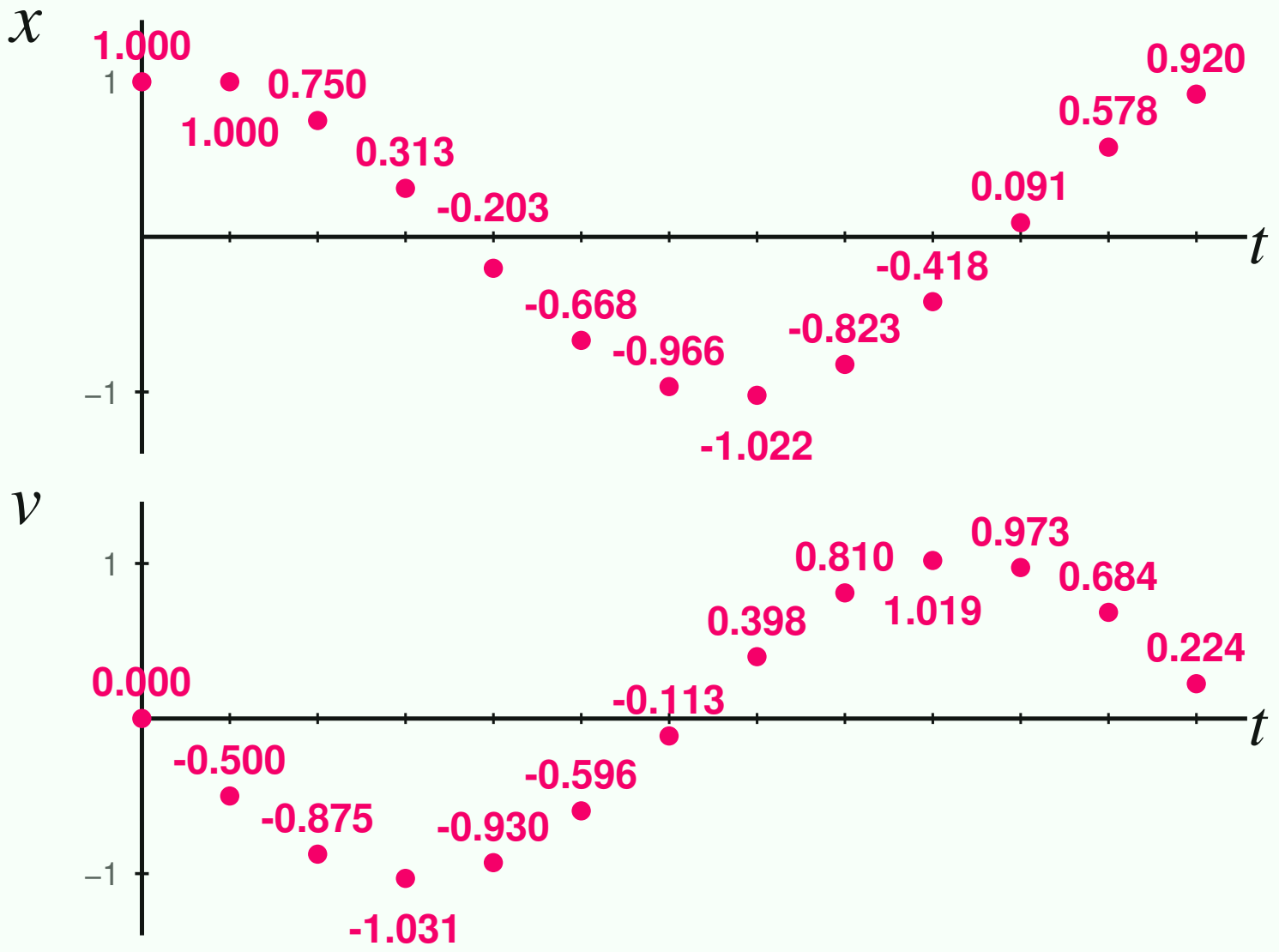
$$\dot{v} = -\frac{k}{m}x$$

k/m = 1 Δt = 0.5

Start: x = 1, v = 0

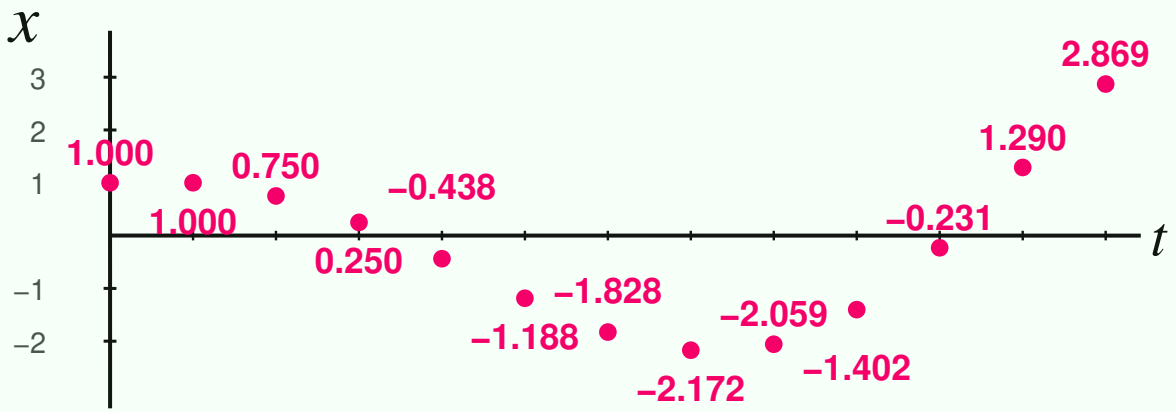
Each step: update x, then v.

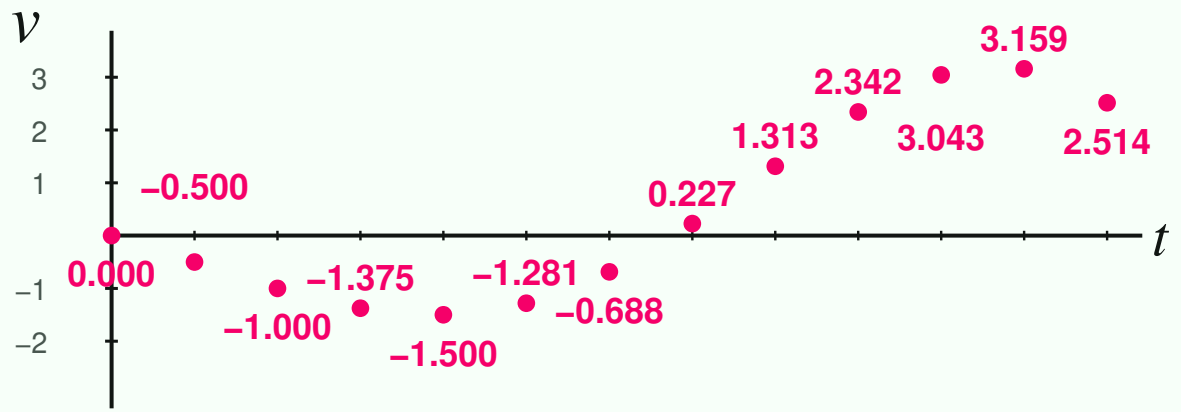
Use the new x when updating v.




```
x = [1, 1.000, 0.750, 0.313, −0.203, −0.668, −0.966, −1.022, −0.823, −0.418, 0.091, 0.578, 0.920]

v = [0, −0.500, −0.875, −1.031, −0.930, −0.596, −0.113, 0.398, 0.810, 1.019, 0.973, 0.684, 0.224]
```





Explicit Euler

k = 1 m = 1 dt = 0.5

x[0] = 1 v[0] = 0

$x[n] = v[n-1] * dt + x[n-1]$

$v[n] = -(k/m)\, x[n-1] * dt + v[n-1]$

n = 12

$2.869 = 3.159 \times .5 + 1.290$

n = 12

$2.514 = -(1/1) \times 1.290 \times .5 + 3.159$

Both calculations use the previous pair: x[n−1], v[n−1].

```
x = [1, 1.000, 0.750, 0.250, -0.438, -1.188, -1.828, -2.172, -2.059, -1.402, -0.231, 1.290, 2.869]

v = [0, -0.500, -1.000, -1.375, -1.500, -1.281, -0.688, 0.227, 1.313, 2.342, 3.043, 3.159, 2.514]
```

Euler’s Explicit Method x[n] = …dt + x[n-1]. It solves differential equations using arithmetic!!!!







Now we will do this in Python!

Software 1.0



